

Lecture 7: Merton Problem with Infinite Horizon

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7.1 Infinite-Horizon Merton Portfolio Choice Problem

In finite-horizon control problems, the investor's value function explicitly depends on calendar time t because the remaining horizon $T - t$ shrinks as time passes. Following the seminal continuous-time framework of Merton [3], modeling an infinite planning horizon ($T \rightarrow \infty$) offers two key structural advantages:

1. **Stationary Policies:** The optimal portfolio weight and consumption rule become time-invariant functions of current wealth x , eliminating calendar time t from the decision rule.
2. **Analytical Tractability:** The temporal derivative vanishes ($\partial_t v = 0$), transforming the dynamic Hamilton-Jacobi-Bellman (HJB) partial differential equation (PDE) into a stationary ordinary differential equation (ODE).

7.1.1 Continuous-Time Market Framework and State-Space Collapse

We consider a frictionless continuous-time economy defined on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions, where the filtration $\{\mathcal{F}_t\}$ is generated by a standard 1D Brownian motion W_t .

The investment universe consists of a risk-free money market account yielding $r > 0$ and a risky asset with price dynamics:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad \text{where } \mu > r \text{ and } \sigma > 0. \quad (7.1)$$

Wealth Process Dynamics: Let $c_t \geq 0$ denote the continuous consumption rate and $\pi_t \in \mathbb{R}$ the total dollar amount allocated to the risky asset at time t (with remaining wealth $X_t - \pi_t$ allocated to the risk-free account). The self-financing wealth evolution dX_t is driven directly by excess returns and net consumption:

$$\begin{aligned} dX_t &= (X_t - \pi_t)r dt + \pi_t \frac{dS_t}{S_t} - c_t dt \\ &= \left(rX_t + \pi_t(\mu - r) - c_t \right) dt + \sigma \pi_t dW_t. \end{aligned} \quad (7.2)$$

State-Space Collapse: Because the percentage asset returns $\frac{dS_t}{S_t}$ are independent of the nominal price level S_t , the asset price S_t drops out as an explicit state variable, leaving scalar wealth $X_t \in \mathbb{R}_+$ as the sole state variable.

Economic Objective: The investor maximizes expected lifetime discounted utility over an infinite horizon:

$$v(x) = \sup_{\{(\pi_t, c_t)\}_{t \geq 0} \in \mathcal{A}} \mathbb{E}_x \left[\int_0^\infty e^{-\rho t} U(c_t) dt \right], \quad (7.3)$$

where $v(x)$ is the stationary value function, $X_0 = x$ is initial wealth, $\rho > 0$ is the subjective discount rate, $U(c)$ is a strictly concave utility function ($U' > 0, U'' < 0$), and $\mathcal{A}$ is the class of admissible, progressive control pairs (π_s, c_s) enforcing non-negative wealth $X_s \geq 0$ almost surely.

7.1.2 Step-by-Step Derivation of the Stationary HJB Equation

We derive the governing stationary equation via the Dynamic Programming Principle (DPP).

Step 1: Time Interval Splitting By the DPP, we split the planning horizon into an immediate interval $[0, \Delta t]$ and the remaining horizon $[\Delta t, \infty)$:

$$v(x) = \sup_{(\pi, c)} \mathbb{E}_x \left[\int_0^{\Delta t} e^{-\rho t} U(c_t) dt + \int_{\Delta t}^\infty e^{-\rho t} U(c_t) dt \right]. \quad (7.4)$$

Step 2: Markov Property and Time Shifting Transforming the second integral via $h = t - \Delta t$ and applying the Law of Iterated Expectations conditionally on $X_{\Delta t}$:

$$\begin{aligned} \mathbb{E}_x \left[\int_{\Delta t}^\infty e^{-\rho t} U(c_t) dt \right] &= e^{-\rho \Delta t} \mathbb{E}_x \left[\mathbb{E}_{X_{\Delta t}} \left[\int_0^\infty e^{-\rho h} U(c_{\Delta t+h}) dh \right] \right] \\ &= e^{-\rho \Delta t} \mathbb{E}_x [v(X_{\Delta t})]. \end{aligned} \quad (7.5)$$

Substituting back into the DPP yields the functional equation:

$$v(x) = \sup_{(\pi, c)_{[0, \Delta t]}} \mathbb{E}_x \left[\int_0^{\Delta t} e^{-\rho t} U(c_t) dt + e^{-\rho \Delta t} v(X_{\Delta t}) \right]. \quad (7.6)$$

Step 3: Asymptotic Expansions as $\Delta t \rightarrow 0^+$ Applying Taylor and Itô expansions to each component:

1. **Flow Utility Integral:** $\int_0^{\Delta t} e^{-\rho s} U(c_s) ds = U(c_0) \Delta t + o(\Delta t)$.
2. **Discount Factor Expansion:** $e^{-\rho \Delta t} = 1 - \rho \Delta t + o(\Delta t)$.
3. **Itô Expansion of $v(X_{\Delta t})$:** For $v(x)$ twice differentiable, Itô's Lemma yields:

$$\mathbb{E}_x [v(X_{\Delta t})] = v(x) + \mathcal{L}^{\pi, c} v(x) \Delta t + o(\Delta t), \quad (7.7)$$

where $\mathcal{L}^{\pi, c}$ is the infinitesimal generator of the wealth process:

$$\mathcal{L}^{\pi, c} v(x) := \left(rx + \pi(\mu - r) - c \right) v'(x) + \frac{1}{2} \sigma^2 \pi^2 v''(x). \quad (7.8)$$

Step 4: Continuous-Time Limit Substituting these expansions into (7.6):

$$v(x) = \sup_{\pi, c} \left\{ U(c_0) \Delta t + (1 - \rho \Delta t) (v(x) + \mathcal{L}^{\pi, c} v(x) \Delta t + o(\Delta t)) \right\}. \quad (7.9)$$

Subtracting $v(x)$, dividing by Δt , and sending $\Delta t \rightarrow 0^+$ yields the **Stationary Hamilton-Jacobi-Bellman (HJB) Equation**:

$$\rho v(x) = \sup_{c \geq 0, \pi \in \mathbb{R}} \left\{ U(c) + \left(rx + \pi(\mu - r) - c \right) v'(x) + \frac{1}{2} \sigma^2 \pi^2 v''(x) \right\}. \quad (7.10)$$

Economic Intuition: The stationary HJB equation equates the required return on the value function ($\rho v(x)$) to the maximum total realized return—the sum of immediate utility flow $U(c)$ and the expected rate of capital growth $\mathcal{L}^{\pi, c}v(x)$.

7.1.3 Explicit Closed-Form Solution under CRRA Utility

We assume Constant Relative Risk Aversion (CRRA) preferences for $c \geq 0$:

$$U(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \text{where } \gamma > 0, \gamma \neq 1. \quad (7.11)$$

Here $\gamma = -cU''(c)/U'(c)$ is the coefficient of relative risk aversion, and $1/\gamma$ is the intertemporal elasticity of substitution (IES).

Step 1: First-Order Conditions (FOCs)

Maximizing the interior expression in (7.10) with respect to c and π :

1. Optimal Consumption c^* :

$$\frac{\partial}{\partial c} (U(c) - cv'(x)) = 0 \implies U'(c^*) = v'(x) \implies c^*(x) = (v'(x))^{-1/\gamma}. \quad (7.12)$$

2. Optimal Risky Allocation π^* :

$$\frac{\partial}{\partial \pi} \left(\pi(\mu - r)v'(x) + \frac{1}{2}\sigma^2\pi^2v''(x) \right) = 0 \implies \pi^*(x) = -\frac{\mu - r}{\sigma^2} \frac{v'(x)}{v''(x)}. \quad (7.13)$$

Step 2: Power Value Function Ansatz

Due to the scale invariance of CRRA utility, we propose the power ansatz:

$$v(x) = A \frac{x^{1-\gamma}}{1-\gamma}, \quad \text{for } A > 0. \quad (7.14)$$

Computing spatial derivatives:

$$v'(x) = Ax^{-\gamma}, \quad v''(x) = -\gamma Ax^{-\gamma-1}. \quad (7.15)$$

Substituting derivatives into the FOCs:

Optimal Portfolio Share Strategy:

$$\pi^*(x) = -\frac{\mu - r}{\sigma^2} \left(\frac{Ax^{-\gamma}}{-\gamma Ax^{-\gamma-1}} \right) = \left(\frac{\mu - r}{\gamma \sigma^2} \right) x. \quad (7.16)$$

The optimal dollar allocation is linear in wealth, yielding a constant portfolio share $\theta^* := \pi^*(x)/x$:

$$\theta^* = \frac{\mu - r}{\gamma \sigma^2}. \quad (7.17)$$

Optimal Consumption Strategy:

$$c^*(x) = (Ax^{-\gamma})^{-1/\gamma} = A^{-1/\gamma}x =: \nu^*x, \quad (7.18)$$

where $\nu^* := A^{-1/\gamma} > 0$ is the constant marginal propensity to consume out of wealth.

Step 3: Determining the Consumption Propensity ν^*

Substituting $\pi^*(x) = \theta^*x$ and $c^*(x) = \nu^*x$ back into the stationary HJB equation (7.10):

$$\rho \left(A \frac{x^{1-\gamma}}{1-\gamma} \right) = \frac{(\nu^*x)^{1-\gamma}}{1-\gamma} + \left(rx + (\theta^*x)(\mu - r) - \nu^*x \right) (Ax^{-\gamma}) + \frac{1}{2} \sigma^2 (\theta^*x)^2 (-\gamma Ax^{-\gamma-1}). \quad (7.19)$$

Using $A = (\nu^*)^{-\gamma}$ and factoring out $x^{1-\gamma}$:

$$\frac{\rho}{1-\gamma} = \frac{\nu^*}{1-\gamma} + r + \theta^*(\mu - r) - \nu^* - \frac{1}{2} \gamma \sigma^2 (\theta^*)^2. \quad (7.20)$$

Substituting $\theta^* = \frac{\mu - r}{\gamma \sigma^2}$:

$$\frac{\rho}{1-\gamma} = \frac{\gamma \nu^*}{1-\gamma} + r + \frac{(\mu - r)^2}{2\gamma \sigma^2}. \quad (7.21)$$

Multiplying by $(1-\gamma)$ and solving for ν^* :

$$\nu^* = \frac{\rho - (1-\gamma)r}{\gamma} - \frac{(1-\gamma)(\mu - r)^2}{2\gamma^2 \sigma^2}. \quad (7.22)$$

7.1.3.1 Step 4: Transversality Condition and Well-Posedness

For the value function to remain finite ($v(x) < \infty$), the optimal propensity to consume must be strictly positive ($\nu^* > 0$). Setting $\nu^* > 0$ in (7.22) yields the **Merton Transversality Condition**:

$$\rho > \rho_{\min} := (1-\gamma) \left(r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right). \quad (7.23)$$

Economic Interpretation:

- **High Risk Aversion** ($\gamma > 1$): Here $1-\gamma < 0$, ensuring $\rho_{\min} < 0$. Since $\rho > 0$, transversality holds universally.
- **Low Risk Aversion** ($\gamma < 1$): Here $1-\gamma > 0$, so $\rho_{\min} > 0$. If $\rho \leq \rho_{\min}$, expected wealth accumulation outpaces the rate of time preference, causing total lifetime utility $\mathbb{E} \left[\int_0^\infty e^{-\rho s} U(c_s) ds \right]$ to explode to $+\infty$.

7.1.4 Summary of Merton Solution Formulas

7.2 Remarks

1. **References:** This lecture is based on the seminal work of Merton [3]. For classic textbook treatments of continuous-time portfolio optimization, see Björk [1]; for rigorous stochastic control foundations, see Fleming and Soner [2].

Variable	Mathematical Formula	Economic Description
Optimal Portfolio Share	$\theta^* = \frac{\mu-r}{\gamma\sigma^2}$	Constant fraction of wealth in risky asset
Optimal Consumption Rate	$c^*(x) = \nu^* x$	Constant propensity to consume out of wealth
Consumption Propensity	$\nu^* = \frac{\rho-(1-\gamma)r}{\gamma} - \frac{(1-\gamma)(\mu-r)^2}{2\gamma^2\sigma^2}$	Annualized consumption-to-wealth fraction
Value Function	$v(x) = \frac{(\nu^*)^{-\gamma}}{1-\gamma} x^{1-\gamma}$	Indirect utility of wealth state x
Transversality Bound	$\rho > (1-\gamma)\left(r + \frac{(\mu-r)^2}{2\gamma\sigma^2}\right)$	Condition preventing utility explosion ($v(x) < \infty$)

References

- [1] T. Björk. *Arbitrage Theory in Continuous Time*. Oxford University Press, Oxford, UK, 4th edition, 2020.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov processes and viscosity solutions*. Springer, 2006.
- [3] R. C. Merton. Lifetime portfolio selection under uncertainty: The continuous-time case. *The review of Economics and Statistics*, pages 247–257, 1969.